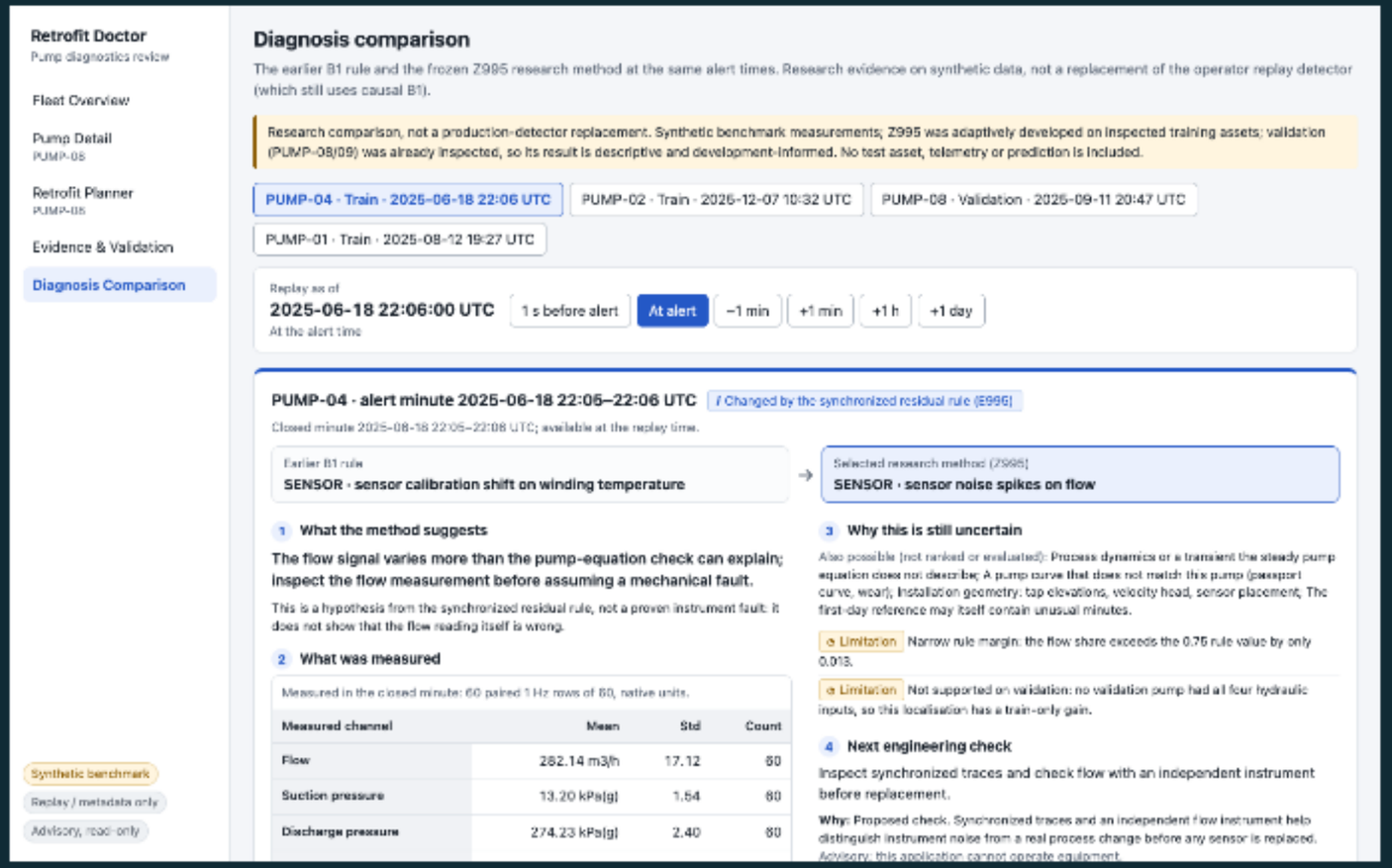
Retrofit Doctor

Pump diagnostics and retrofit review

Synthetic benchmark demonstration
Since AI 2026



Twelve independent pump assets

Split	Assets	Demo access
Training	PUMP-01–07	Development and selection
Validation	PUMP-08–09	Labeled replay and evaluation
Test	PUMP-10–12	Equipment metadata only

Different sensor coverage and observation weeks.
Site grouping does not imply hydraulic coupling.

Diagnostic gains and a remaining error

Case / split	Earlier B1	Combined method
PUMP-04 / train	Calibration shift, winding temperature	Flow-sensor noise Correct type and channel
PUMP-08 / validation	Unknown cause	Bearing wear Correct fault type
PUMP-01 / train	Suction-pressure calibration shift	Same incorrect cause Flow check unavailable

Engineering checks still determine the cause.

Retrospective synthetic labels. Field confirmation remains open.

Diagnosis improved on training data

Metric	Causal B1	Combined method
Correct fault type	8 / 26	10 / 26
Sensor fault type + channel	3 / 8	4 / 8
Events detected	18 / 26	18 / 26
False + repeated alert episodes	6	6

Seven pumps, 26 events and 49 asset-days. Synthetic data.

Two corrected causes. No loss of earlier correct diagnoses.
New independent data is needed to establish generalization.

Validation: 4/9 to 5/9 correct types, from vibration. Previously inspected data.

Sensor-information test with the baseline

Three training pumps, 10 events. Same causal B1 detector.

Information available	Detected / 10	Correct type / 10	False + excess
All installed groups	9	2	6
Without vibration / bearing temperature	8	0	6
Without suction pressure	9	2	6
Without flow / power	8	3	5

Restoring each group reproduces the original result.

Vibration / temperature improved overall counts.
Other effects were mixed. Installation benefit remains unproven.

Synthetic information ablation. No physical installation benefit or ROI established.

Proposed retrofit architecture

Existing measurements first, targeted additions where needed

Element	Proposed implementation	Site decision needed
Existing signals	Pressure, flow and VFD speed, where installed	Tag mapping, time alignment, update cadence
Missing measurement	Targeted flow or housing vibration / temperature	Suitability, mounting, range and measurement bandwidth
Gateway	Read-only collection, timestamps, local buffering and advisory analysis	IO, power, environment, maintenance owner
Engineering display	Evidence, uncertainty and next engineering check	Review workflow, reference checks

The demo replays synthetic data. The proposed gateway needs site validation.
Existing controls and operational alarms remain independent.

No hardware built. No selected supplier, fixed BOM or certification claim.

A retrofit must justify its full cost

PUMP-08 has no flow channel. Flow assessment is a hypothesis:
pipe suitability, reference checks and diagnostic value remain unverified.

Illustrative redundant-pump case	EUR
Initial incremental installation cost	4,500
Annual benefit / nuisance checks / upkeep	300 / 1,800 / 700
Annual net effect	−2,200
Five-year incremental value (0% discount)	−15,500

Defer permanent retrofit under these assumptions.

Hypothetical standby case; not this pump’s topology, a quote or measured savings.

Pilot pathway

01 Observe existing data in shadow mode

02 Check whether warnings support useful action

03 Trial the smallest measurement that resolves a gap

04 Verify installation, integration and maintenance

05 Decide using evidence, full cost and nuisance burden

Proposed pilot. Benefits, physical feasibility and applicable approvals remain unvalidated.